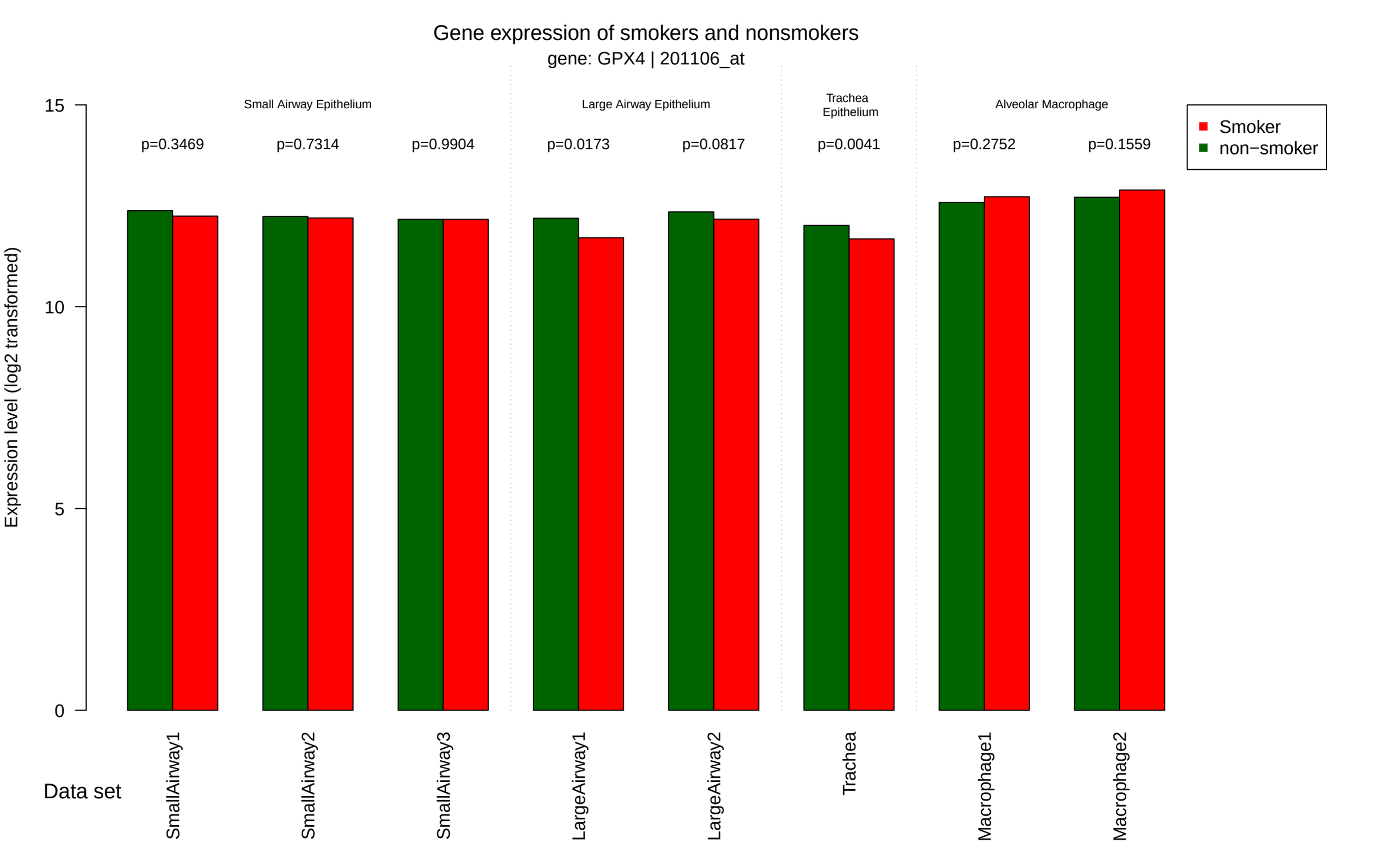
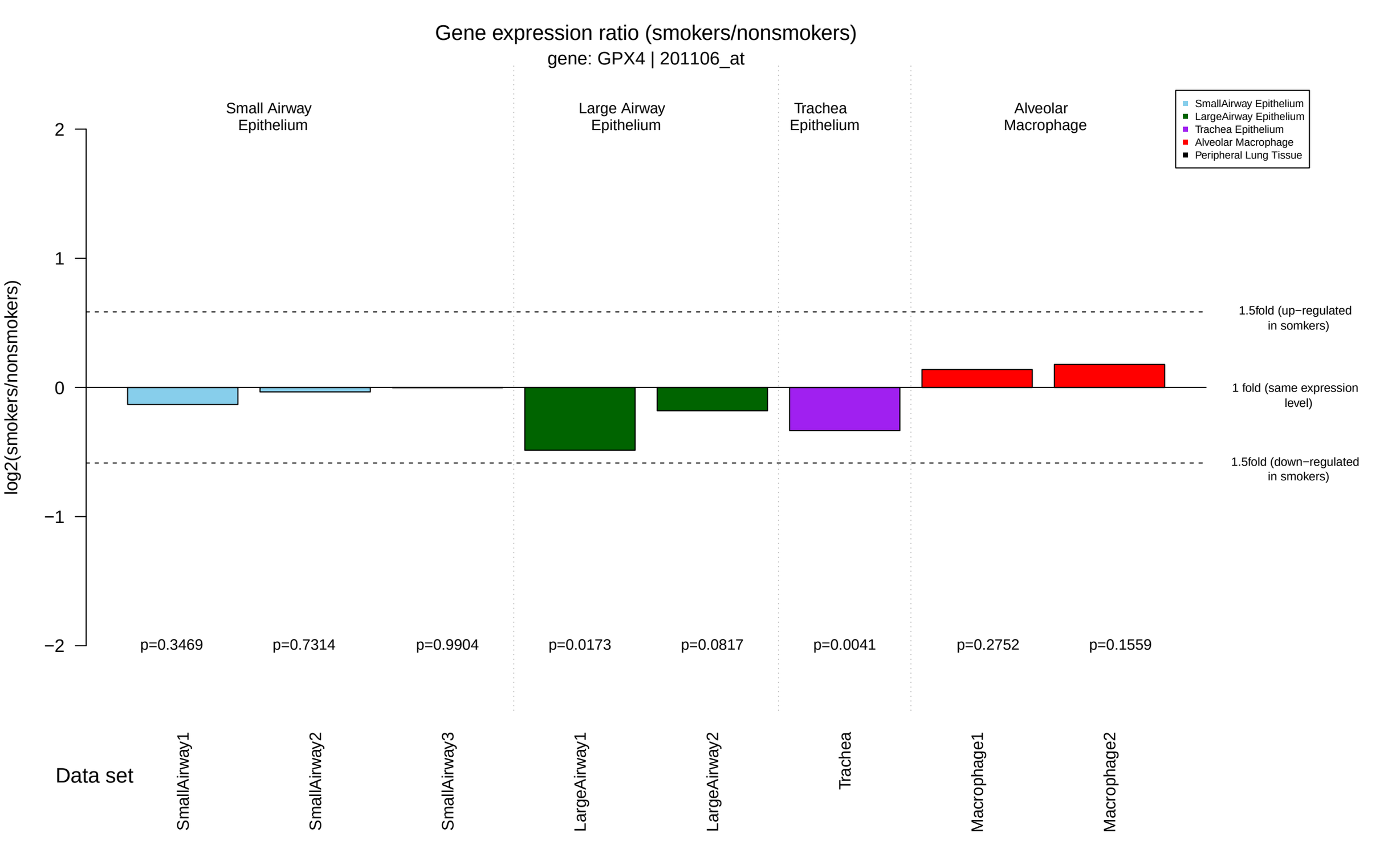
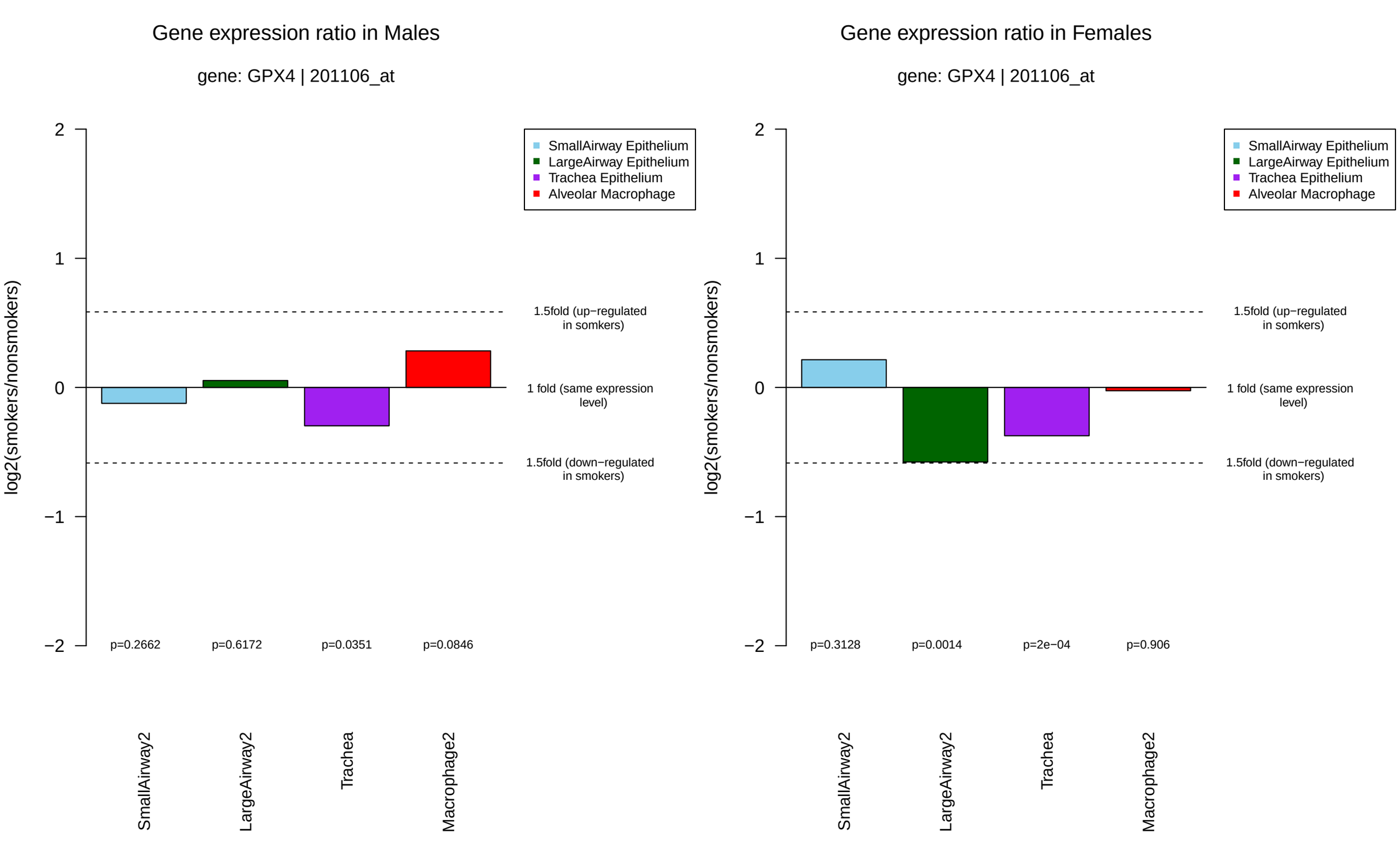
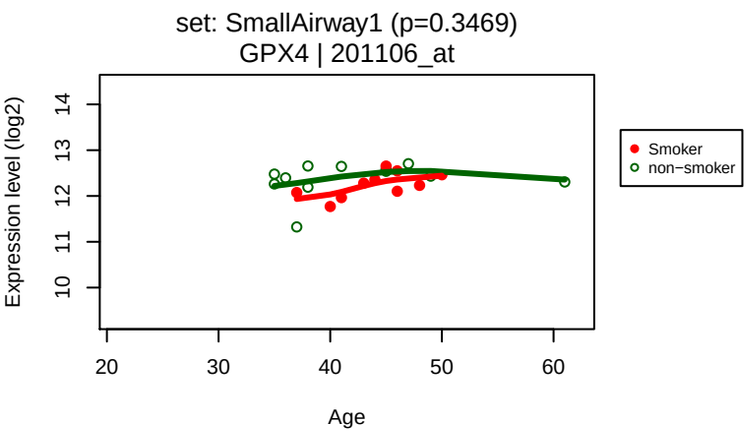
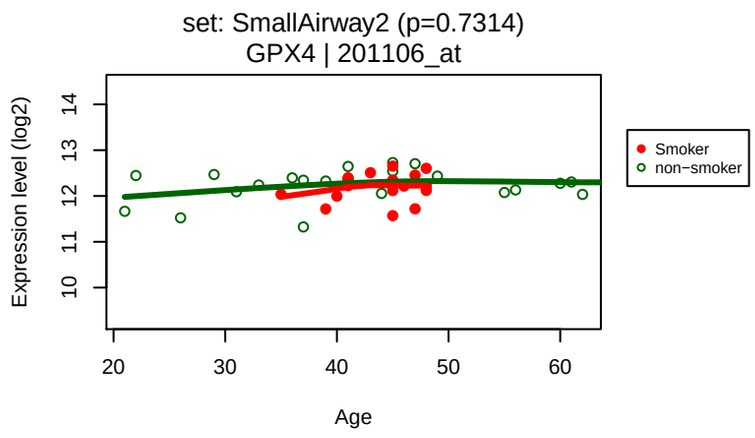
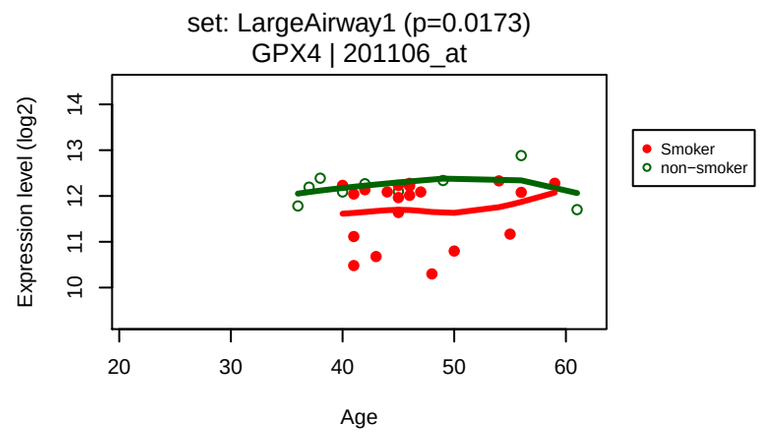

Gene expression of smokers and nonsmokers
gene: GPX4 | 201106_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
p=0.3469
p=0.7314
p=0.9904
p=0.0173
p=0.0817
p=0.0041
p=0.2752
p=0.1559
Smoker
non−smoker
Expression level (log2 transformed)
15
10
5
0
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

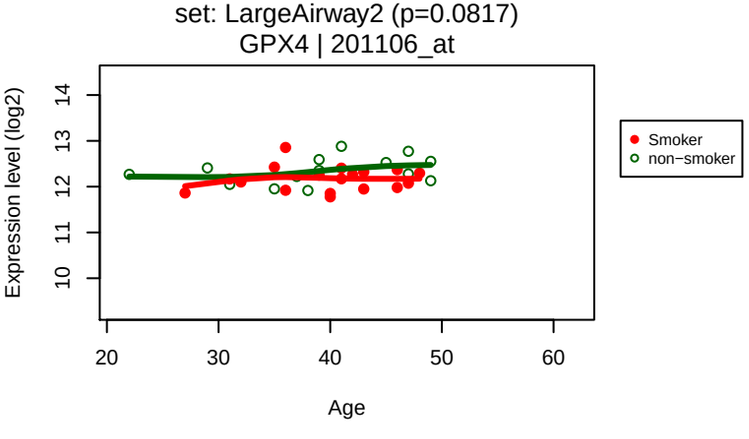

Gene expression ratio (smokers/nonsmokers)
gene: GPX4 | 201106_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Peripheral Lung Tissue
log2(smokers/nonsmokers)
2
1
0
−1
−2
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.3469
p=0.7314
p=0.9904
p=0.0173
p=0.0817
p=0.0041
p=0.2752
p=0.1559
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

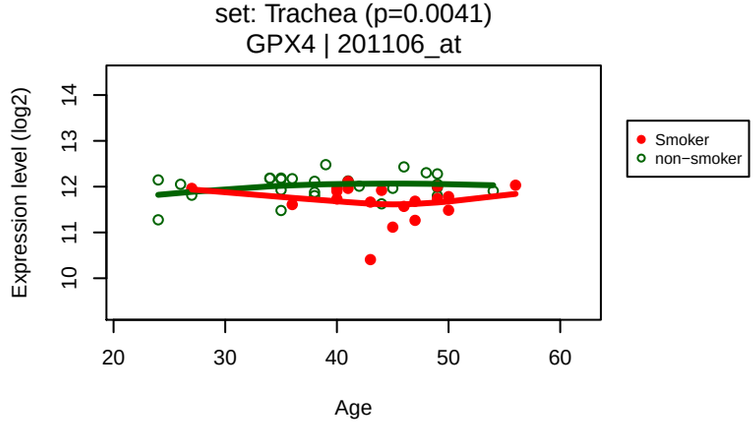

Gene expression ratio in Males
gene: GPX4 | 201106_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.2662
p=0.6172
p=0.0351
p=0.0846
SmallAirway2
LargeAirway2
Trachea
Macrophage2
Gene expression ratio in Females
gene: GPX4 | 201106_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.3128
p=0.0014
p=2e−04
p=0.906
SmallAirway2
LargeAirway2
Trachea
Macrophage2

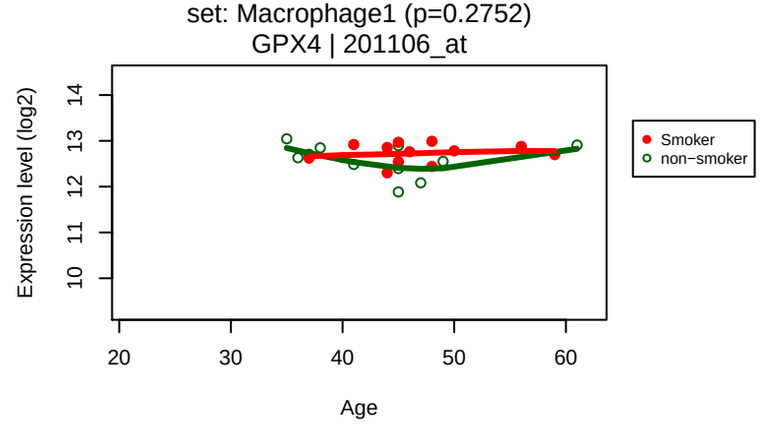

set: SmallAirway1 (p=0.3469)
GPX4 | 201106_at
Expression level (log2)
Age
Smoker
non−smoker

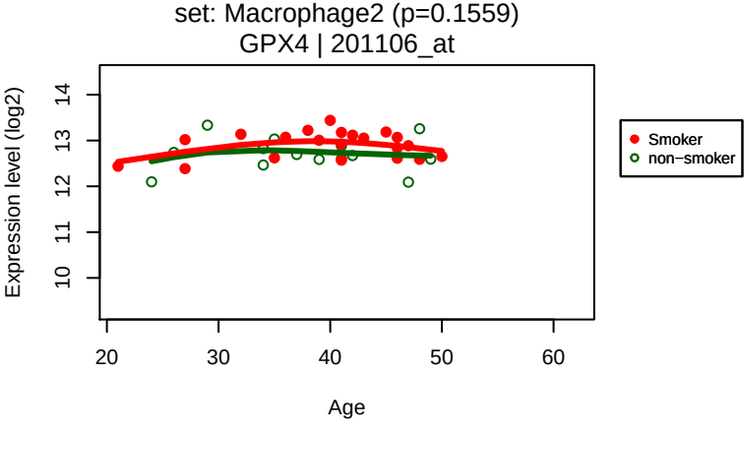

set: SmallAirway2 (p=0.7314)
GPX4 | 201106_at
Expression level (log2)
Age
Smoker
non−smoker

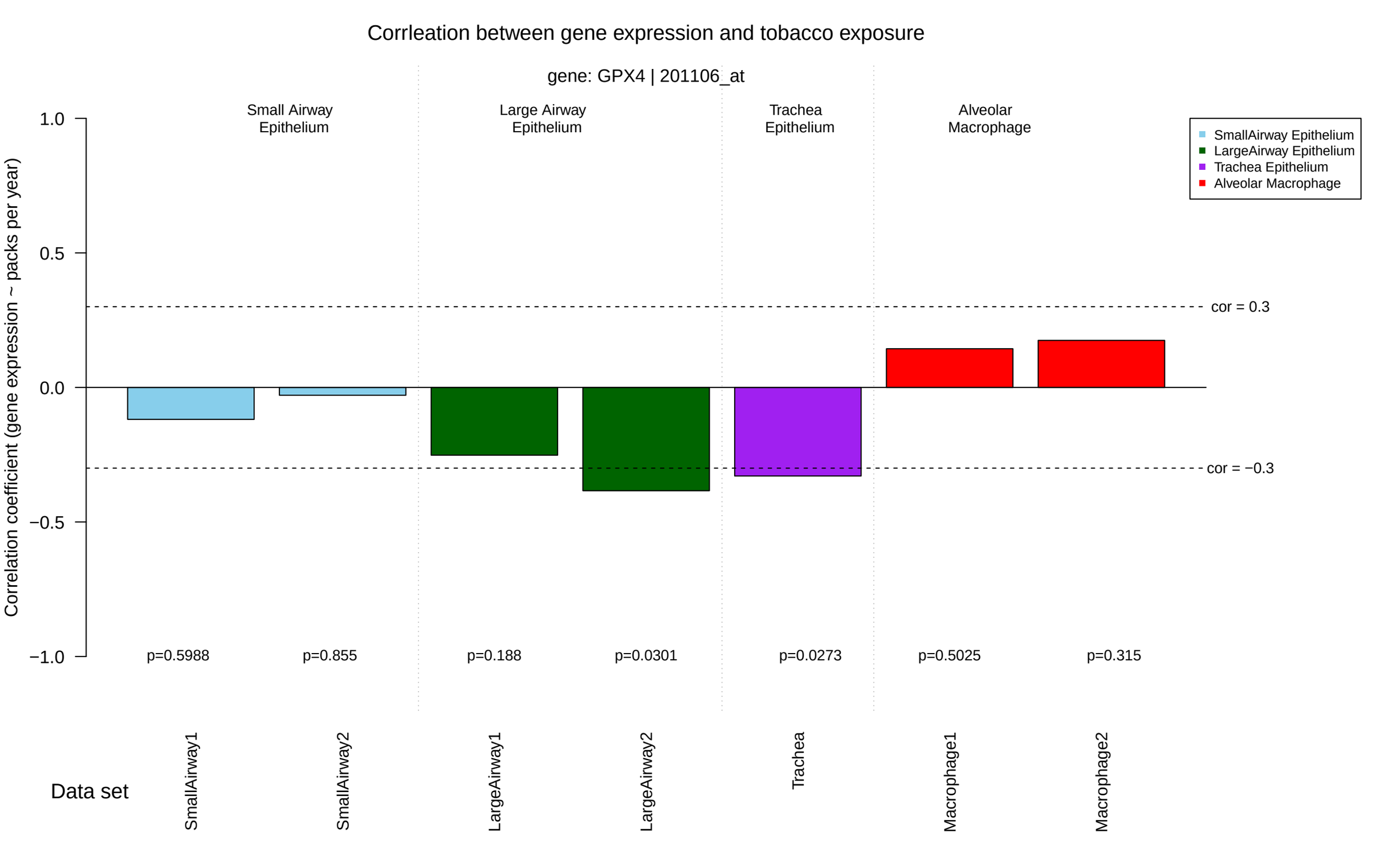

set: LargeAirway1 (p=0.0173)
GPX4 | 201106_at
Expression level (log2)
Age
Smoker
non−smoker


set: LargeAirway2 (p=0.0817)
GPX4 | 201106_at
Expression level (log2)
Age
Smoker
non−smoker


set: Trachea (p=0.0041)
GPX4 | 201106_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage1 (p=0.2752)
GPX4 | 201106_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage2 (p=0.1559)
GPX4 | 201106_at
Expression level (log2)
Age
Smoker
non−smoker


Corrleation between gene expression and tobacco exposure
gene: GPX4 | 201106_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Correlation coefficient (gene expression ~ packs per year)
1.0
0.5
0.0
−0.5
−1.0
cor = 0.3
cor = −0.3
p=0.5988
p=0.855
p=0.188
p=0.0301
p=0.0273
p=0.5025
p=0.315
Data set
SmallAirway1
SmallAirway2
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2